

Sana Anand

Bioengineering undergraduate skilled in CAD, MATLAB, PCR, & prototyping with a strong interest in mechanics.

EDUCATION

University of California, San Diego

San Diego, CA

B.S. in Bioengineering | GPA: 3.65

Expected: June 2028

Relevant Coursework: Mechanical Design, Bioengineering, MATLAB, Linear Circuits (Robotics), Organic Chemistry

SKILLS

Engineering & Design: AutoCAD (Certified), Fusion 360 (Certified), SolidWorks (Beginner), Manufacturing Processes, Statics, Circuit Analysis, GD&T, 3D Printing

Programming & Tools: MATLAB, Arduino (Beginner), Microsoft Office/Excel, Raspberry Pi

Biomedical & Lab: PCR, Gel Electrophoresis, Sample Preparation, Animal Handling, CPR/BLS (Certified)

Languages: English (Native), Spanish (Conversational)

PROJECTS

Electromechanical Robot - Course Project

- Applied introductory manufacturing techniques to fabricate a scissor-lift robot component, including laser cutting, drilling, and mechanical assembly using hand tools.
- Integrated electrical functionality by soldering and wiring DC motors, validated through functional testing.

Pendulum Clock - Course Project

- Designed and constructed a pendulum clock, applying physics principles and mechanical assembly techniques.

RELEVANT EXPERIENCE

Rady Children's Hospital - Hematology/Oncology

San Diego, CA

Shadowing

August 2025 – September 2025

- Studied common pediatric solid tumors and their treatment strategies, including diagnostic testing, chemotherapy delivery systems, and anesthesia workflows and observed clinical implementation of treatment protocols.

UC San Diego

San Diego, CA

Lab Assistant

January 2025 – March 2025

- Performed DNA extraction and PCR analysis to identify genetic mutations related to cardiac disease research.
- Conducted animal handling and sample collection following lab safety and ethical protocols.
- Supported postdoctoral researchers and PI with experimental setup, data organization, and lab maintenance.

Americare Neurosurgery International (AMCANI)

Orinda, CA

Intern

May 2023 – May 2024

- Supported global neurosurgical medical missions by coordinating logistics and outreach initiatives.
- Managed and optimized organizational website to communicate technical medical outcomes and mission impact.
- Led fundraising and digital campaigns to secure resources for medical education and surgical programs in under-resourced regions.

ACTIVITIES

Engineering World Health | Engineering Team Member

Sep. 2024 – Sep. 2025

- Designed and prototyped a low-cost prosthetic arm using CAD (AutoCAD, Fusion 360), 3D printing, & Arduino with a focus on improving accessibility for under-resourced communities.

AMSA Health Fair | Event Organizer

Sep. 2024 – Present

- Organize a health fair for the San Diego community by coordinating mobile clinics and healthcare vendors to provide free medical resources, streamlined through my translation between English and Spanish.

Camp Kesem | Counselor

Sep. 2025 – Present

- Served as a counselor providing mentorship and structured activities for children affected by a parent's cancer.
- Fundraise to ensure campers could attend programming free by engaging community members and to support Camp Kesem's mission.